

ACADIANA, LA, USA

WMO: 722314

Lat: 30.038N

Lon: 91.884W

Elev: 24

StdP: 14.68

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 53915

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.2	34.0	17.9	13.5	38.4	22.5	17.0	42.0	24.9	58.2	23.4	56.2	7.5	0	0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.6	94.7	77.9	92.9	77.7	91.2	77.4	81.1	88.3	80.3	87.7	79.7	87.1	6.4	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.3	152.3	83.5	78.8	149.6	83.4	77.5	143.3	83.1	44.6	88.2	43.7	87.7	43.0	86.8	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.9	19.1	17.2	25.1	97.9	3.5	2.8	22.5	100.0	20.5	101.6	18.5	103.2	16.0	105.3		
			22.9	82.3	3.5	1.0	20.3	83.0	18.3	83.6	16.3	84.1	13.7	84.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	69.4	52.9	56.9	63.0	69.3	76.2	81.3	82.6	82.9	79.5	70.8	61.2	54.9
	DBStd	12.77	9.94	9.39	7.87	6.22	4.82	2.90	2.40	2.65	4.05	7.12	8.43	9.46
	HDD50	197	85	37	8	0	0	0	0	0	0	0	12	55
	HDD65	1345	391	253	132	32	2	0	0	0	0	30	169	336
	CDD50	7267	174	231	411	581	813	940	1011	1020	886	644	348	208
	CDD65	2940	15	27	70	163	351	490	546	555	436	209	54	24
	CDH74	27202	32	79	336	1149	3111	4946	5710	5838	3883	1712	323	83
	CDH80	10534	0	2	23	192	1074	2131	2433	2575	1555	514	33	2
Wind	WSAvg	7.3	8.9	9.3	9.0	9.3	7.8	6.1	4.7	4.8	5.8	6.3	7.3	8.5
Precipitation	PrecAvg	61.5	5.6	4.4	3.9	5.0	5.0	6.8	6.0	5.7	5.3	4.7	4.3	4.8
	PrecMax	85.3	13.3	9.1	10.6	11.9	13.5	17.9	9.3	27.9	10.3	13.9	13.8	13.5
	PrecMin	40.4	0.8	0.7	0.3	0.5	0.3	0.9	1.8	1.5	2.0	0.2	1.2	2.3
	PrecStd	11.3	3.6	2.4	2.4	2.9	3.4	3.7	1.9	4.8	2.4	3.9	3.0	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.2	79.4	82.3	86.4	91.9	97.0	96.2	97.7	94.8	90.4	83.2	79.4
		MCWB	68.7	69.4	68.8	72.0	73.8	76.9	78.4	78.4	77.5	74.8	71.6	70.6
	2%	DB	74.5	76.5	80.4	84.0	89.8	93.1	93.5	94.8	92.3	88.0	80.8	76.5
		MCWB	67.2	68.2	68.7	71.4	74.0	76.8	78.4	78.3	76.9	74.7	70.3	69.8
	5%	DB	71.9	73.5	78.3	82.1	88.0	91.1	91.6	92.8	90.4	85.4	78.3	73.2
		MCWB	66.1	67.0	68.1	70.2	74.1	76.8	78.4	78.4	76.4	73.0	68.5	68.0
	10%	DB	68.8	71.7	75.1	80.7	85.9	89.8	90.4	90.9	88.4	82.5	75.0	70.5
		MCWB	64.4	66.0	66.7	69.4	73.5	76.9	78.2	78.2	75.7	71.7	66.8	66.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.2	71.9	73.4	75.9	78.3	81.2	81.8	82.3	80.9	79.5	75.2	73.9
		MCDB	74.7	76.3	78.4	82.4	86.3	88.1	89.1	89.5	86.8	84.7	79.5	77.0
	2%	WB	69.4	70.3	71.9	74.5	77.1	80.1	80.9	81.3	79.7	77.9	73.1	71.4
		MCDB	72.6	74.4	76.3	80.4	84.7	87.4	88.4	88.6	86.6	83.8	77.3	74.8
	5%	WB	67.3	68.9	70.4	73.4	76.2	79.1	80.1	80.4	78.7	76.3	71.2	69.5
		MCDB	70.4	72.7	74.9	78.7	83.5	86.7	87.7	87.8	85.9	82.2	75.5	73.0
	10%	WB	64.8	67.0	69.0	72.3	75.4	78.3	79.4	79.7	77.8	74.8	69.1	66.9
		MCDB	67.9	70.3	73.8	77.7	82.4	85.9	87.0	87.2	85.1	80.1	74.2	69.9
Mean Daily Temperature Range	5% DB	MDBR	19.0	18.4	19.3	19.2	17.9	16.2	15.8	16.6	17.3	20.7	21.1	18.9
		MCDBR	19.2	18.6	18.4	18.9	18.6	18.3	17.5	18.7	18.8	20.4	21.2	19.0
		MCWBR	13.4	11.8	9.8	8.9	6.9	6.1	6.3	6.4	6.7	9.2	11.9	13.3
	5% WB	MCDBR	16.9	15.2	15.8	15.7	15.6	15.7	16.0	16.5	16.0	16.6	17.8	17.6
		MCWBR	13.8	11.1	10.5	9.2	6.7	6.5	6.6	6.7	6.6	9.0	12.7	14.0
Clear-Sky Solar Irradiance	taub	0.339	0.356	0.379	0.420	0.448	0.473	0.492	0.485	0.445	0.381	0.359	0.351	
	taud	2.503	2.465	2.389	2.277	2.237	2.212	2.193	2.209	2.290	2.444	2.454	2.474	
	Ebn at Noon	284	288	288	278	269	261	256	256	264	276	273	272	
	Edh at Noon	29	32	37	43	45	46	47	45	40	32	30	28	
All-Sky Solar Radiation	RadAvg	929	1089	1485	1807	1990	1949	1852	1750	1613	1429	1077	832	
	RadStd	91	157	133	131	114	157	134	154	143	163	103	84	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+220	+144

Nomenclature: See separate page